

Superfecta Project Report

A clean, report-safe read on what actually improved, what deserves paper-trade trust now, and what still does *not* deserve hype.

Phase 7 = primary paper baseline

Phase 8 = shadow challenger

Selective rules = only paper-worthy family

Main conclusion: the project got meaningfully better through **validation cleanup, decision clarity**, and **operational hardening**, not through fake “overnight edge” claims.

BEST CLEAN HISTORICAL GAIN

+7.96 pts

Historical selector-scoring improvement: walk-forward ROI improved from **+22.46%** to **+30.42%** with sqrt-dampened scoring, but the current frozen selector benchmark still remains the **Train-only yearly selector** at **+22.46%** walk-forward and **+14.36%** holdout.

PRIMARY BASELINE

Phase 7 OP/CD rule-component basket • 2024 +0.37% on 109 •
2025 +105.38% on 66

+38.68%

SHADOW CHALLENGER

Phase 8 frozen portfolio • 2024 +9.50% on 85 • 2025 +50.26%
on 33

+21.45%

LIVE PEAK TODAY

Best demo EV seen

+23.24%

ALERT BAR

Watcher threshold

+30.00%

HOLDOUT WINNER

+38.68%

Phase 7 holdout ROI on **175 races**, currently the strongest practical paper baseline, but with an uneven recent path: **2024 +0.37% on 109, 2025 +105.38% on 66.**

HISTORICAL SELECTOR UPGRADE

+30.42%

Historical selector-scoring result after cleanup. The current frozen selector benchmark still remains **+22.46%** walk-forward and **+14.36%** holdout for the **Train-only yearly selector**.

OPERATIONAL SHIFT

Paperable

Daily paper-trade, settlement, forward-check, lane monitor, and preflight note path are now in place.

That is workflow hardening, not new forward evidence by itself.

LIVE STATUS

No +30 yet

Demo/watcher lane is working, but today's best live EV stayed below the real alert threshold.

Master findings register

This is the one-section synthesis of the entire project state, combining the strongest findings from the strategy, validation, portfolio, model, operations, and live-demo lanes.

CATEGORY	CORE FINDING	EVIDENCE
Portfolio hierarchy	Phase 7 is still the best current deployment baseline.	Phase 7: +38.68% on 175 (2024 +0.37% on 109; 2025 +105.38%
Selector quality	Validation improved materially after selector cleanup, but the historical scoring win is not the same thing as the current frozen benchmark.	Historical scoring: +22.46% → +30.42%; current benchmark: +2
Rule stack	OP_DURABLE_K7 remains the safest anchor, with CD_CORE_K8 still paper-worthy.	Forward trust scorecard + holdout / WF comparisons
Method families	Selective rules are the only family that earned current-paper credibility.	Selective path positive; Harville -24.05%; ML betting line -24.16%

CATEGORY	CORE FINDING	EVIDENCE
ML / features	Horse-history features improved prediction quality, but not enough to change downstream betting behavior.	Better RMSE / R^2 , negligible EV-engine pass-through change
Full-data retrain caveat	Full-data XGBoost retrain artifacts are command and diagnostic reproducibility only.	FULL_DATA_RETRAIN_ARTIFACTS.md + validate_full_data_retrain
Operations	The paper-trade and reporting stack is now operationally real.	Settlement sync, helper, forward checks, lane monitors, preflight
Current paper bridge	Current paper totals must route through the source-checked current-evidence bridge before they are quoted.	CURRENT_EVIDENCE_SUMMARY.md / current_evidence_summary
Live / demo lane	The live scanner works, but today	Best seen today +23.24%; watcher threshold +30.00%

CATEGORY	CORE FINDING	EVIDENCE
	still has no honest +30% edge.	
PRIMARY CONCLUSION Phase 7 still owns the room The project’s prettiest headline is no longer the thing that matters most. The strongest practical evidence still belongs to the Phase 7 OP/CD rule-component basket, not the bigger Phase 8 expansion • Best current holdout: +38.68% on 175 races • Recent split: 2024 +0.37% on 109, 2025 +105.38% on 66 — real edge, but not a smooth two-year path • Best choice for the main paper-trade lane right now		
VALIDATION UPGRADE Selector cleanup was worth it The most defensible quantitative improvement in the whole project came from cleaning the selection logic, not from inventing a new strategy family, but it should still be read as a historical scoring improvement rather than the current frozen deployment benchmark. • Historical selector-scoring result improved from +22.46% to +30.42% • Current frozen selector benchmark still remains the Train-only yearly selector at +22.46% walk-forward and +14.36% holdout • No new rule mining, no fake re-ranking of history		
RULE HIERARCHY Trust is now easier to explain The project finally has a clearer answer to the question “which rules do we actually trust?” instead of spreading that answer across ten files. The year split is what makes the hierarchy easier to defend honestly.		

- **OP_DURABLE_K7** = safest anchor: mixed holdout, but on the largest OP holdout sample (**2024 -47.41% on 68; 2025 +124.61% on 47**) plus the strongest walk-forward support (**7/10 folds**)
- **CD_CORE_K8** = paper-worthy, still useful in the primary paper lane: the steadier current paper candidate because it stayed positive in both holdout years (**2024 +45.65% on 41; 2025 +78.21% on 19**)
- **OP_REFINED_K7** = interesting, but still a challenger not a replacement: prettier aggregate ROI, but still a smaller mixed-year spike (**2024 -25.47% on 33; 2025 +210.02% on 16**)

So the real read is not “highest ROI wins.” It is that CD currently looks steadier, OP_DURABLE still has the stronger anchor-grade evidence base, and OP_REFINED still needs more forward sample before it can challenge the anchor seriously.

REJECTED HYPE

Some improvements were real, but not promotable

One of the best things about the current report is that it explicitly rejects seductive but weak stories instead of pretending every metric bump deserves deployment.

- Harville remains a benchmark, not a live system
- XGBoost residual correction remains research, not a betting engine
- Full-data XGBoost retrain commands and RMSE / MAE gains stay in `FULL_DATA_RETRAIN_ARTIFACTS.md / validate_full_data_retrain_artifacts.py` as model-fit diagnostics only
- Horse-history features helped prediction, but not enough downstream to change the bet path

OPERATIONAL MATURITY

This is no longer just a research pile

The paper-trade system now behaves like something you can actually run and interpret day to day, instead of a loose set of scripts with unclear status.

- Primary + shadow lanes
- Settlement sync and entry helper
- Forward-check and lane-monitor artifacts
- Preflight note for “OP/CD not racing today” cases

Quote the bridge before the totals

For short Cole updates, read `CURRENT_EVIDENCE_SUMMARY.md` / `current_evidence_summary.json` before quoting `PAPER_TRADE_NOW`, ops-bucket/operator-status context, settlement-audit, or primary-ledger totals.

- Source consistency is **matched** across the top card, audit, and primary settlement CSV.
- Combined operator read route says check `operator_status_context`, `source_freshness.requires_refresh_before_right_now_use=false`, and `operator_read_gate.requires_refresh_before_evidence_read=false`; the saved `PAPER_TRADE_NOW` best-action card is fresh against the bridge reference date but still goes through operator read-gate routing before right-now instruction or evidence use. This is not no-target, clean empty, bet-readiness, settled-ROI, promotion, live-profitability, bankroll, or real-money evidence.
- Gate source is `forward_evidence_scorecard.json` `decision_gate_minimums`: **anchor_displacement=30, phase8_promotion_review=20, real_money_discussion=100**; these are future ROI-complete observation floors, not cleared gates.
- Scorecard audit route: `current_evidence_summary.json` source path `scorecard_audit_route` points copied gate/ranking/CI-only/timezone/no-BAQ synchronization checks to `SCORECARD_RANKING_CONTRACT_AUDIT.md` / `scorecard_ranking_contract_audit.json` plus `python3 validate_scorecard_ranking_contract_audit.py`; this route is report-synchronization metadata only, not forward performance, settled ROI, promotion readiness, live profitability, bankroll guidance, or real-money evidence.
- Rebuild-order route: `current_evidence_summary.json` source path `rebuild_validation_contract` routes scorecard/rules/signals/settlement-ledger byte change through `python3 paper_trade_settlement_audit.py -> python3 current_evidence_summary.py -> python3 validate_current_evidence_summary.py` before quoting `CURRENT_EVIDENCE_SUMMARY.*`; this is provenance/rebuild metadata only, not settled ROI, promotion readiness, live profitability, bankroll guidance, or real-money evidence.
- Primary paper is still **6/30** ROI-complete toward a first read and **6/100** toward broader review.
- Current settled sample is **CD-only: CD_CORE_K8 6, OP_DURABLE_K7 0**; this is not OP-anchor proof, promotion readiness, live profitability, bankroll guidance, real-money evidence, or cross-family promotion evidence. For anchor / paper / watch caveats, read `CROSS_FAMILY_DECISION.md` and run `python3 validate_cross_family_decision.py`; stale-card refresh routing, CD-only settled rows, source-published settlement-queue state/context and green HTML/PDF validation are not OP-anchor proof.
- OP_REFINED CI-only check: Scorecard CI-only promotion check: `forward_evidence_scorecard.json:ci_only_promotion_diagnostics.OP_REFINED_K`

keeps `ci_only_promotion_allowed=false`; OP_REFINED's positive CI lower bound is support context only, not a current-paper promotion trigger, OP-anchor proof, promotion readiness, live profitability, bankroll guidance, or real-money evidence.

PDF bridge text: Combined operator read route says check `operator_status_context`, `source_freshness.requires_refresh_before_right_now_use=false`, and `operator_read_gate.requires_refresh_before_evidence_read=false`; the saved PAPER_TRADE_NOW best-action card is fresh against the bridge reference date but still goes through operator read-gate routing before right-now instruction or evidence use. This is not no-target, clean-empty, bet-readiness, settled-ROI, promotion, live-profitability, bankroll, or real-money evidence.

Gate source is `forward_evidence_scorecard.json` `decision_gate_minimums`: `anchor_displacement=30`, `phase8_promotion_review=20`, `real_money_discussion=100`; these are future ROI-complete observation floors, not cleared gates.

If `source_consistency.overall_match=false`, repair the top-card / audit / CSV mismatch before quoting current paper numbers. Then use the combined `operator_status_context` + `source_freshness.requires_refresh_before_right_now_use=false` + `operator_read_gate.requires_refresh_before_evidence_read=false` route before using the saved right-now card as current-day guidance or evidence.

LIVE REALITY

Working lane, disciplined trigger

The demo lane is real now, and that matters. But the discipline matters just as much: a functioning scanner is not the same thing as a live edge.

- Best live number seen today: **+23.24%**
- Alert threshold remains **+30.00%**
- Correct behavior is still to stay quiet until the number actually earns it

WHAT ACTUALLY GOT BETTER

The biggest real improvements

This repo is stronger now because the evidence is cleaner, the operating path is real, and the live/demo lane no longer gets confused with production.

1

Validation got sharper

The strongest clean historical numeric gain is the selector-scoring improvement from **+22.46%** to **+30.42%**, using frozen artifacts and no new rule mining, while the current frozen selector benchmark still remains the **Train-only yearly selector**.

2

Portfolio hierarchy got clearer

Phase 7 remains the live paper baseline. Phase 8 stays useful, but only as a shadow challenger until it beats Phase 7 on better forward evidence. The current edge is real, but the year split shows it was uneven rather than smooth.

3

Operations got real

The project now has daily paper-trade runners, settlement tooling, forward checks, lane monitors, and preflight context for quiet race days.

BEFORE

Strong but messy

Good research, too much drift, too many files saying half of the story.

NOW

Defensible + usable

Cleaner deployment logic, clearer ranking of what is paper / shadow / benchmark / skip.

NET EFFECT

Less bullshit

Better odds of saying the true thing quickly, especially under pressure or while showing work to Cole.

QUANTIFIED CHANGE

Numbers that matter

These are the metrics worth saying out loud. They hold up better than the older shiny-but-looser headline stats.

Historical selector-scoring ROI

+22.46% → +30.42%



Phase 7 holdout ROI

+38.68% on 175 races • 2024 +0.37% on 109 • 2025 +105.38% on 66



Phase 8 holdout ROI

+21.45% on 118 races • 2024 +9.50% on 85 • 2025 +50.26% on 33



Train-only yearly selector holdout ROI

+14.36% on 65 races



The cleanest sentence in the whole project is still this: **Phase 7 beats Phase 8 on the current holdout, even though Phase 8 looks prettier on paper — but the winning Phase 7 path was uneven, not a smooth two-year glide.**

What to run, what to watch, what to cite

The live choice, the shadow choice, and the benchmark choice are not the same thing anymore.
That’s good.

APPROACH	ROLE	HOLDOUT
Phase 7 OP/CD rule- component basket	PAPER NOW	+38.68% • 175 races • 2024 +0.37% on 109 • 2025 +105.38% on 109
Phase 8 frozen portfolio	SHADOW ONLY	+21.45% • 118 races • 2024 +9.50% on 85 • 2025 +50.26% on 85

APPROACH	ROLE	HOLDOUT
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Train-only
yearly
selector

BENCHMARK

+14.36% • 65 races

PRIMARY CALL

Phase 7 first

Use the Phase 7 OP/CD rule-component basket as the real paper-trade default, with target cards confirmed by daily preflight, because it owns the best current holdout evidence.

CHALLENGE LANE

Phase 8 in shadow

Log it, compare it, but do not promote it just because it has more moving pieces or a prettier walk-forward headline.

Which family deserves trust?

This is intentionally separate from the selective-method ranking. First rule out dead-end families, then compare the serious selective contenders against each other.

METHOD FAMILY	STATUS	BEST HONEST EVIDENCE
Selective rule path	PAPER NOW	+38.68% holdout • 2024 +0.37% on 109 • 2025 +105.38%
Harville-ranked probabilities	BENCHMARK ONLY	-24.05% ROI • 90,004 races
XGBoost residual	RESEARCH ONLY	-24.16% ML betting ROI • 16,724 races

METHOD FAMILY	STATUS	BEST HONEST EVIDENCE
correction		
The broader selective-family secondary lines elsewhere in the repo are replay context on walk-forward test years, not extra train-only validation.		

RULE STACK

Current rule hierarchy

These are the rules worth naming first when somebody asks what the project actually trusts.

RULE	TIER	HOLDOUT	WF
OP_DURABLE_K7	ANCHOR	+22.9% • 115	7/10 selected
CD_CORE_K8	PAPER	+56.0% • 60	1/10 selected
OP_REFINED_K7	WATCH	+51.4% • 49	2/10 selected

OPERATIONAL HARDENING

Why the project feels more real now

The strongest practical gain is not a headline stat, it's that the daily operating path stopped being hand-wavy. That makes the workflow more reproducible, not newly profitable by itself.

DAILY

Primary + shadow runners

Separate paper lanes make it obvious what is the real baseline versus what is only being watched.

AFTER RACES

Settlement sync + helper

There is now a clean place to record actual outcomes and returns instead of hand-editing ledgers and hoping nothing drifts.

VALIDATION

Forward checks + lane monitors

Forward evidence can now be read as *no data*, *too early*, *running cold*, or *within noise* instead of being guessed from raw CSVs.

QUIET DAYS

Preflight note

Quiet days now explain whether OP/CD simply were not racing, instead of making the project look broken or low-signal by accident.

Today's live reality

The live model lane works. The watcher works. The threshold discipline is the point.

WORKING NOW

Separate demo lane

`demo\_live\_predictions.py` now handles real available cards without pretending the production OP/CD basket changed.

- Best-live selection mode
- PLAY / FLOW / PASS labeling
- Candidate leaderboard output
- Cache hardening + latest JSON snapshots
- Silent watcher unless a true +30% EV spot appears

TODAY'S BEST

Laurel R9 • +23.24%

Encouraging, but still below the alert bar. That is why the watcher stays quiet.



WHAT STILL DOES NOT CLEAR

Do not over-claim these

This project got better partly because some tempting stories were explicitly rejected.



Do not say Phase 8 replaced Phase 7.

It didn't. Phase 8 is still shadow-only until it wins on better forward evidence.



Do not say ML fixed the betting case.

Horse-history features improved prediction quality, but downstream EV decisions barely changed.



Do not claim a +30 live opportunity happened today.

Best seen was +23.24%, which is below the real alert bar.



Do say the project is much more defensible now.

The report hierarchy, paper-trade path, and live/demo separation are all materially better than before.

FINAL STANCE

The clean recommendation

If somebody only remembers one paragraph from this report, it should be this one.

The project improved meaningfully, but in the right way: better validation, better decision hygiene, better paper-trade operations, and a working live/demo lane that does not fake production readiness. The strongest current deployment stance is still conservative: paper trade the selective rule path, keep Phase 7 as the primary baseline, keep Phase 8 as shadow-only, and keep Harville/XGBoost out of the live betting decision path.

Built from the current Superfecta evidence stack on [2026-04-15](#).

Core inputs: compare_main_approaches, forward_evidence_scorecard, portfolio decision card, method-family decision card, selector experiment, sample-size experiment, overnight progress, and the live demo watcher outputs.